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Math 201 Final Exam Term 221

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If D is the triangular region with vertices $(0,1)$, $(1,2)$, $(4,1)$, then $\iint_D y^2\,dA =$

(a) $\dfrac{13}{3}$

(b) $\dfrac{7}{3}$

(c) $\dfrac{11}{3}$

(d) $\dfrac{17}{3}$

(e) $\dfrac{9}{3}$

2. $\int_{-2}^{2}\int_{0}^{\sqrt{4-y^2}}\int_{\frac{x^2+y^2}{4}}^{\frac{\sqrt{x^2+y^2}}{2}} xz\,dz\,dx\,dy =$

(a) $\dfrac{1}{16}\int_0^2 \left(4r^4-r^6\right)dr$

(b) $\int_0^1 \left(r^3-r^5\right)dr$

(c) $\int_0^1 \left(r^4-r^6\right)dr$

(d) $\dfrac{1}{32}\int_0^1 \left(4r^4-r^6\right)dr$

(e) $\dfrac{1}{16}\int_0^2 \left(4r^3-r^5\right)dr$

3. The exact length of the curve $x = 1+3t^2,\ y = 4+2t^3,\ 0 \le t \le 1$ is

(a) $6\sqrt{2}-1$

(b) $\sqrt{2}+1$

(c) $\sqrt{2}-3$

(d) $4\sqrt{2}-2$

(e) $\sqrt{2}+3$

4. $\int_0^1 \int_{x^2}^1 \sqrt{y} \sin y \, dy \, dx =$

(a) $\int_0^1 \int_0^y \sqrt{y} \sin y \, dx \, dy$

(b) $\int_0^1 \int_0^{2y} \sqrt{y} \sin y \, dx \, dy$

(c) $\int_0^1 \int_0^{\sqrt{y}} \sqrt{y} \sin y \, dx \, dy$

(d) $\int_0^1 \int_y^{\sqrt{y}} \sqrt{y} \sin y \, dx \, dy$

(e) $\int_0^1 \int_0^{y^2} \sqrt{y} \sin y \, dx \, dy$

5. The value of $\iiint_E z \, dV$, where E is the solid bounded by the coordinate planes and the plane $x + y + z = 1$ is

(a) $\frac{1}{2}$

(b) $\frac{1}{6}$

(c) $\frac{1}{24}$

(d) $\frac{1}{48}$

(e) $\frac{1}{12}$

6. The graph of $r \cot \theta = 6 \sec \theta$ is

(a) an ellipse

(b) a parabola

(c) a straight line

(d) a circle

(e) a hyperbola

7. The surface $\rho = \sin \theta \sin \phi$ represents

(a) a sphere with radius $\frac{1}{4}$ and center $\left(0, \frac{1}{2}, 0\right)$

(b) a sphere with radius $\frac{1}{4}$ and center $(0, 1, 0)$

(c) a sphere with radius $\frac{1}{2}$ and center $(0, 1, 0)$

(d) a sphere with radius $\frac{1}{2}$ and center $\left(0, \frac{1}{2}, 0\right)$

(e) a sphere with radius 1 and center $(0, 0, 0)$

8. If $w = xy$, $x = \cos t$ and $y = \sin t$, then $\dfrac{dw}{dt} =$

(a) 1

(b) $2\cos 2t$

(c) $\cos 2t$

(d) $\sin 2t$

(e) $2\sin 2t$

9. Let $R = [0, 4] \times [-1, 2]$. Use a Riemann sum with $m = 2$ (number of subintervals in x-direction) and $n = 3$ (number of subintervals in y-direction) to estimate $\iint_R (2 - 2xy^2)\, dA$. Take the sample points to be upper left corners of the rectangles.

(a) -18

(b) -12

(c) -16

(d) -14

(e) -10

10. The quadratic surface $x^2 - y^2 + z^2 - 4x + 2y - 4z + 6 = 0$ represents

(a) a paraboloid

(b) a hyperboloid of one sheet

(c) an ellipsoid

(d) a cone

(e) a hyperboloid of two sheets

11. The value of k for which the vectors $\vec{a} = \langle 1, 2, 3\rangle$, $\vec{b} = \langle 0, 1, 1\rangle$ and $\vec{c} = \langle k, 0, -1\rangle$ lie in the same plane is

(a) -2

(b) 2

(c) -1

(d) 0

(e) 1

12. The sum of the extreme values of the function $f(x, y) = xe^y$, subject to the constraint $x^2+y^2 = 2$ is

(a) 0

(b) 1

(c) e

(d) $-e$

(e) $2e$

13. Let E be the solid lying above the cone $z = \sqrt{x^2 + y^2}$ and between the spheres $x^2+y^2+z^2 = 1$ and $x^2 + y^2 + z^2 = 4$. Then $\iiint_E \sqrt{x^2 + y^2 + z^2}\, dV =$

(a) $\dfrac{15\pi}{2}\left(1 - \sqrt{3}\right)$

(b) $\dfrac{15}{4}\pi\left(1 + \dfrac{\sqrt{2}}{2}\right)$

(c) $\dfrac{15}{2}\pi\left(1 + \dfrac{\sqrt{2}}{2}\right)$

(d) $\dfrac{15}{2}\pi\left(1 - \dfrac{\sqrt{2}}{2}\right)$

(e) $\dfrac{15}{4}\pi\left(1 - \dfrac{\sqrt{2}}{2}\right)$

14. The function $f(x, y) = x^4 - 2x^2 + y^3 - 3y$ has how many critical points?

(a) 8

(b) 3

(c) 5

(d) 4

(e) 6

15. If B is the ball with center at the origin and radius 2, then $\iiint_B \left(x^2 + y^2 + z^2\right)^2 dV =$

(a) $\dfrac{506\pi}{7}$

(b) $\dfrac{510\pi}{7}$

(c) $\dfrac{508\pi}{7}$

(d) $\dfrac{512\pi}{7}$

(e) $\dfrac{500\pi}{7}$

16. The maximum rate of change of $f(x, y) = xe^y$ at the point $P(2, 0)$ is

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) 0

(d) $\sqrt{5}$

(e) 1

17. If $\vec{a} = \langle -2, 3, 1\rangle$ and $\vec{b} = \langle 1, 2, 2\rangle$, then $\text{comp}_{\vec{a}}\, \vec{b} =$

(a) $\dfrac{6}{\sqrt{14}}$

(b) $\dfrac{4}{\sqrt{14}}$

(c) $\dfrac{8}{\sqrt{14}}$

(d) $\dfrac{10}{\sqrt{14}}$

(e) $\dfrac{3}{\sqrt{14}}$

18. The absolute maximum of $f(x, y) = e^{-x^2-y^2}\left(2x^2 + y^2\right)$ on the disk $x^2 + y^2 \leq 4$ is

(a) $\dfrac{4}{e}$

(b) $\dfrac{4}{e^2}$

(c) $\dfrac{1}{e}$

(d) $\dfrac{2}{e^2}$

(e) $\dfrac{2}{e}$

19. The volume of the region enclosed by the cylinder $x^2 + (y-1)^2 = 1$ and the planes $z = 0$, $z = y$ is

(a) $\int_0^{2\pi} \int_0^{\sin\theta} \int_0^{\cos\theta} r\, dz\, dr\, d\theta$

(b) $\int_0^{\pi} \int_0^{\cos\theta} \int_0^{r\sin\theta} r\, dz\, dr\, d\theta$

(c) $\int_0^{\pi} \int_0^{2\sin\theta} \int_0^{r\sin\theta} r\, dz\, dr\, d\theta$

(d) $\int_0^{\pi/2} \int_0^{\sin\theta} \int_0^{r\cos\theta} r\, dz\, dr\, d\theta$

(e) $\int_0^{\pi} \int_0^{2\cos\theta} \int_0^{r\sin\theta} r\, dz\, dr\, d\theta$

20. The value of $\iint_D e^{-y^2}\,dA$, where $D = \{(x, y) : 0 \le y \le 2,\ 0 \le x \le y\}$ is equal to

(a) $\frac{1}{2}\left(1 - e^{-3}\right)$

(b) $\frac{1}{2}\left(1 - e^{-6}\right)$

(c) $\frac{1}{2}\left(1 - e^{-4}\right)$

(d) $\frac{1}{2}\left(1 - e\right)$

(e) $\frac{1}{2}\left(1 - e^{-2}\right)$

21. If

$$\iiint_E x\,dV = \int_0^a \int_0^b \left(cr^2 - r^4\right)\cos\theta\,dr\,d\theta,$$

where E is the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$, then $a + b - c =$

(a) $\pi - 2$

(b) 2π

(c) $4(\pi - 1)$

(d) $2(\pi - 1)$

(e) $\pi - 4$

22. The volume of the solid bounded by the plane $z = 0$ and the paraboloid $z = 1 - x^2 - y^2$ is (Hint: Use polar coordinates)

(a) $\frac{\pi}{2}$

(b) π

(c) 2π

(d) $\frac{\pi}{4}$

(e) 0

23. The iterated integral $\int_1^2 \int_0^3 \left(6x^2y - 4x\right)dy\,dx$ is equal to

(a) 45

(b) 21

(c) -52

(d) 48

(e) 42

24. If $f(x, y) = x^y$, then $f_{xy}(x, y)$ at $x = 1$ and $y = 2$ is

(a) 3

(b) 2

(c) 1

(d) 5

(e) 4

25. The distance between the parallel planes $10x + 2y + 2z = 10$ and $5x + y + z = 2$ is

(a) 1

(b) $\dfrac{1}{\sqrt{2}}$

(c) $\dfrac{1}{\sqrt{3}}$

(d) $\sqrt{3}$

(e) $\sqrt{2}$

26. The function $f(x, y) = xy + \dfrac{1}{x} + \dfrac{1}{y}$ has local minimum value at $(x, y) =$

(a) $(-1, 0)$

(b) $(0, 0)$

(c) $(-1, -1)$

(d) $(-1, 1)$

(e) $(1, 1)$

27. Let E be the region bounded by the cone $z^2 = x^2 + y^2$ and the paraboloid $z = 2 - x^2 - y^2$. Then $\iiint_E z\, dV =$

(a) π

(b) $\dfrac{5\pi}{4}$

(c) $\dfrac{11\pi}{12}$

(d) $\dfrac{4\pi}{5}$

(e) 2π

28. The value of the integral

$$I = \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} (2x + y)\, dx\, dy$$

is

(a) $\frac{5}{3}$

(b) $\frac{1}{3}$

(c) $\frac{2}{3}$

(d) $\frac{8}{3}$

(e) $\frac{4}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$\int_1^2 \int_{y-1}^{7-3y} y^2 dx\, dy = \int_1^2 (8y^2 - 4y^3) dy = \frac{11}{3}$.
2	**(a)**	Cylindrical, half disk: $\frac{\cos\theta}{32}(4r^4 - r^6)$, $\int \cos = 2$: $\frac{1}{16}\int_0^2 (4r^4 - r^6) dr$.
3	**(d)**	$L = \int_0^1 6t\sqrt{1+t^2}\, dt = 2(2\sqrt{2} - 1) = 4\sqrt{2} - 2$.
4	**(c)**	Region $x^2 \le y \le 1$: for fixed y, $0 \le x \le \sqrt{y}$, $0 \le y \le 1$.
5	**(c)**	$\int_0^1 \int_0^{1-x} \int_0^{1-x-y} z\, dz\, dy\, dx = \frac{1}{6} \cdot \frac{1}{4} = \frac{1}{24}$.
6	**(b)**	$r\frac{x}{y} = \frac{6r}{x} \Rightarrow x^2 = 6y$, a parabola.
7	**(d)**	$\rho^2 = y$: $x^2 + (y - \frac{1}{2})^2 + z^2 = \frac{1}{4}$, radius $\frac{1}{2}$, center $(0, \frac{1}{2}, 0)$.
8	**(c)**	$\frac{dw}{dt} = -\sin^2 t + \cos^2 t = \cos 2t$.
9	**(c)**	$\Delta A = 2$; sum of f at the six upper left corners is -8; estimate -16.
10	**(b)**	Completing squares: $(x-2)^2 - (y-1)^2 + (z-2)^2 = 1$, one sheet.
11	**(c)**	$\det = -1 - k = 0 \Rightarrow k = -1$.
12	**(a)**	$y = x^2$, $x = \pm 1$: $f = e$ and $f = -e$, sum 0.
13	**(d)**	Spherical: $\frac{15}{4} \cdot 2\pi \cdot \left(1 - \frac{\sqrt{2}}{2}\right) = \frac{15\pi}{2}\left(1 - \frac{\sqrt{2}}{2}\right)$.
14	**(e)**	$x \in \{0, \pm 1\}$, $y \in \{\pm 1\}$: $3 \cdot 2 = 6$ critical points.

Q	Ans	Brief explanation
15	**(d)**	$\int_0^2 \rho^6 d\rho \cdot 2\pi \cdot 2 = \frac{128}{7} \cdot 4\pi = \frac{512\pi}{7}$.
16	**(d)**	$\lvert\nabla f(2,0)\rvert = \lvert\langle 1, 2\rangle\rvert = \sqrt{5}$.
17	**(a)**	$\frac{\vec{a}\cdot\vec{b}}{\lvert\vec{a}\rvert} = \frac{6}{\sqrt{14}}$.
18	**(e)**	Interior critical points give $f(\pm 1, 0) = \frac{2}{e}$; boundary maximum is $\frac{8}{e^4} < \frac{2}{e}$.
19	**(c)**	$r = 2\sin\theta$, $0 \le \theta \le \pi$, $0 \le z \le r\sin\theta$.
20	**(c)**	$\int_0^2 y e^{-y^2} dy = \frac{1}{2}(1 - e^{-4})$.
21	**(d)**	Cylindrical: $a = 2\pi$, $b = 2$, $c = 4$, so $a + b - c = 2(\pi - 1)$.
22	**(a)**	$V = \int_0^{2\pi} \int_0^1 (1 - r^2) r \, dr \, d\theta = 2\pi \cdot \frac{1}{4} = \frac{\pi}{2}$.
23	**(a)**	Inner: $27x^2 - 12x$; outer: $[9x^3 - 6x^2]_1^2 = 48 - 3 = 45$.
24	**(c)**	$f_{xy} = x^{y-1} + yx^{y-1}\ln x$, which equals 1 at $(1, 2)$.
25	**(c)**	Point $(1, 0, 0)$: $D = \frac{\lvert 5-2\rvert}{\sqrt{27}} = \frac{1}{\sqrt{3}}$.
26	**(e)**	Only critical point $(1, 1)$; $D = 3 > 0$, $f_{xx} = 2 > 0$: local minimum.
27	**(c)**	$2\pi \int_0^1 \frac{r}{2}[(2 - r^2)^2 - r^2] dr = \frac{11\pi}{12}$.
28	**(c)**	Polar over the upper half disk: $I = \frac{1}{3} \cdot 2 = \frac{2}{3}$.